Composite Terms in Sparse Cubic Pisot Floors: Alternating Shifts, 3-Adic Limits, and Projective Return

Keio

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Abstract

For a cubic Pisot number β, we study the integer sequence $\lfloor \beta^{3^r} \rfloor$. We prove that this sequence has infinitely many composite terms whenever the two non-dominant conjugates are non-real. For totally real cubic Pisot numbers we obtain coefficient obstructions, finite quadratic-character cycle criteria, and projective finite-field return criteria. These imply positive lower density of composite indices for several explicit infinite families. A counterexample shows that an individual prime term need not divide any later term. We also give a family with prime first terms but an eventual fixed-divisor progression, including a completely certified three-prime prefix. The results address proper parts of Saito's Problem 1.1; the totally real case in full, and the transcendence of Mills' constant, are not proved. The general trace congruences and finite-field facts are classical; the contribution examined here is their combination for the original sparse floor sequence.

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1 Introduction and exact scope

A Pisot number is a real algebraic integer $\beta > 1$ all of whose other conjugates have modulus less than one. Saito asked whether every cubic Pisot number satisfies

$$\#\{r \geq 0 : \left\lfloor \beta^{3^r} \right\rfloor \text{ is composite}\} = \infty. \tag{1}$$

This is Problem 1.1 in the arXiv version of [1]. Removing or adding finitely many initial indices does not change the question. Existing results for compositions with reversible recurrence sequences do not automatically apply to the exponent 3^r.

The first main result is the following.

Theorem 1.1 (Non-real cubic case; [source T45]). *If β is a cubic Pisot number with non-real conjugates, then (1) holds.*

The second group of results concerns totally real β. Write its minimal polynomial as

$$f(X) = X^3 - aX^2 + bX - c \in \mathbb{Z}[X], \tag{2}$$

with roots β, u, v, and put $s = 1$ if $u + v > 0$ and $s = 0$ if $u + v < 0$.

Theorem 1.2 (Coefficient obstructions; [source T34,36,50]). *If $s = 0$ and $c \not\equiv ab \pmod 3$, then (1) holds. If $s = 1$ and $6 \nmid c$, the same conclusion holds. More precisely, eventual primality would force*

$$L_f := \lim_{r\to\infty} \mathrm{Tr}(\beta^{3^r}) = s + \varepsilon \quad \text{in } \mathbb{Z}_3 \qquad (\varepsilon \in \{-1, 1\}).$$

For either sign ε,

$$L_f = \varepsilon \iff a \equiv \varepsilon \pmod 3, \qquad c \equiv ab \pmod 3.$$

The third group consists of criteria producing arithmetic progressions of composite indices. They use divisors of a trace term, rather than requiring that the term itself be prime. In the zero-shift case, returning the ratios of the reduced conjugates suffices; the individual conjugates need not return.

All proofs below are independent of the other eight manuscripts in the proposed series. Source identifiers refer to the author's supplied research audit and are retained only to make the editorial correspondence transparent. No claim of priority over all prior literature is made. The limitations are mathematical as well as bibliographical: even the union of our real criteria is not the full set of real cubic Pisot numbers.

2 Trace identities, shifts, and congruences

For the three roots $\alpha_1, \alpha_2, \alpha_3$ of (2), define

$$T_r = \sum_i \alpha_i^{3^r}, \qquad B_r = \sum_{i<j} (\alpha_i\alpha_j)^{3^r}, \qquad E_r = c^{3^r}.$$

These are integers, with initial state $(T_0, B_0, E_0) = (a, b, c)$. Newton's identities give the exact polynomial map

$$\Phi(T, B, E) = (T^3 - 3BT + 3E,\ B^3 - 3EBT + 3E^2,\ E^3). \tag{3}$$

In particular $T_r \equiv a \pmod 3$ for every r.

Lemma 2.1 (Power fields)**.** *For every positive integer m, a Pisot number β satisfies $\mathbb{Q}(\beta^m) = \mathbb{Q}(\beta)$. Consequently its powered conjugates are distinct.*

Proof. Every embedding of $\mathbb{Q}(\beta)$ over $\mathbb{Q}(\beta^m)$ sends β to a conjugate whose mth power is β^m. Such a conjugate has modulus $\beta > 1$, and hence is β itself. Characteristic zero extensions are separable, so the relative degree is one. Distinctness follows by extending embeddings from the equal fields. □

For real cubic roots, $|u| = |v|$ would mean $u = -v$, making $\beta = a \in \mathbb{Z}$, contrary to degree three. Thus $u + v \neq 0$, and for all sufficiently large r,

$$\left\lfloor \beta^{3^r} \right\rceil = T_r - s. \tag{4}$$

Indeed, the sum of the two small odd powers has the sign of the larger small root, which is also the sign of $u + v$, and tends to zero.

Lemma 2.2 (Ghost congruence)**.** *For every $r \geq 0$,*

$$T_{r+1} - T_r \in 3^{r+1}\mathbb{Z}. \tag{5}$$

Consequently $L_f = \lim_r T_r$ exists in $\mathbb{Z}_3$ and

$$L_f - T_k \in 3^{k+1}\mathbb{Z}_3 \qquad (k \geq 0). \tag{6}$$

Proof. Let A be an integral companion matrix of f. For any integral matrix and any prime ℓ, write its trace of the ℓ^jth power as a sum over closed words of length ℓ^j in its matrix indices. Cyclic rotation groups such words into orbits of length $1, \ell, \ldots, \ell^j$. A word obtained by repeating a primitive word of length ℓ^i contributes an integer weight raised to ℓ^{j-i}. Hence there are integers w (the weights of primitive cyclic words) such that

$$\operatorname{Tr}(A^{\ell^j}) = \sum_{i=0}^{j} \ell^i \sum_{w \in \mathcal{W}_i} w^{\ell^{j-i}}.$$

The new terms with $i = j$ are divisible by ℓ^j. In the difference between the expressions for j and $j - 1$, every older term is also divisible by ℓ^j, since $w^{\ell^h} - w^{\ell^{h-1}}$ is divisible by ℓ^h for integral w. The latter follows from Fermat's congruence and induction, using the binomial theorem (including $\ell \mid w$). Set $\ell = 3$ and $j = r + 1$. Summing the resulting congruences proves (6) and the Cauchy assertion. □

This is a standard trace/Witt congruence, not a new ingredient by itself.

Lemma 2.3 (Affine finite-field return)**.** *Let $p \nmid 3c \operatorname{disc}(f)$ be prime. Let $d \in \{1, 2, 3\}$ be the splitting-field degree of $f \bmod p$, and suppose*

$$p \mid T_r - s, \qquad v_3(p^d - 1) \leq r.$$

Then there is an explicitly computable $K \geq 1$ such that

$$p \mid T_{r+jK} - s \qquad (j \geq 0).$$

If (4) holds eventually, all sufficiently late members of this progression are composite.

Proof. Write $p^d - 1 = 3^t M$, where $(M, 3) = 1$ and $t \leq r$. Take $K = \operatorname{ord}_M(3)$, setting $K = 1$ for $M = 1$. Then $p^d - 1$ divides $3^r(3^{jK} - 1)$. Each of the three nonzero roots in $\mathbb{F}_{p^d}$ therefore has the same power at indices $r + jK$ and r. Their sums return as well. Since β^{3^n} tends to infinity, eventually the divisible floor exceeds p. □

Whenever shifts have a fixed period h, the same proof works after replacing K by hK. This observation will be essential in the non-real case.

3 The non-real case

Write the small conjugates as $\alpha = \rho e^{i\theta}$ and $\bar{\alpha}$, with $0 < \rho < 1$. By Lemma 2.1, $\alpha/\bar{\alpha}$ is not a root of unity. In particular $\cos(3^r\theta)$ never vanishes. For large r define

$$s_r = \mathbf{1}_{\{\cos(3^r\theta)>0\}}, \qquad q_r = \left\lfloor \beta^{3^r} \right\rfloor = T_r - s_r.$$

Lemma 3.1. *Both values $s_r = 0$ and $s_r = 1$ occur infinitely often.*

Proof. If the cosine and its next triple-angle cosine are both positive, the formula $\cos 3x = \cos x(4\cos^2 x - 3)$ forces $\cos x > \sqrt{3}/2$; if both are negative it forces $\cos x < -\sqrt{3}/2$. Thus an eventually constant sign would put every sufficiently late angle within $\pi/6$ of 0 or, respectively, π modulo 2π. Write these angles as a fixed centre plus δ_r, where $|\delta_r| < \pi/6$. Tripling admits no branch change inside these intervals, so $\delta_{r+1} = 3\delta_r$. Boundedness for all future r implies $\delta_r = 0$. This would make $(\alpha/\bar{\alpha})^{3^r} = 1$, a contradiction. □

Suppose, seeking a contradiction, that all sufficiently late q_r are prime. They are then distinct, odd, and different from 3. If $a \equiv 0 \pmod 3$, the infinitely many indices with $s_r = 0$ give multiples of 3; if $a \equiv 1 \pmod 3$, use $s_r = 1$. Hence

$$a \equiv 2 \pmod 3. \tag{7}$$

Modulo 2, the map (3) becomes

$$(T, B, E) \longmapsto (T + BT + E,\ B + EBT + E,\ E).$$

Its complete state graph is the following eight-state list:

$$000 \mapsto 000, \quad 001 \mapsto 111, \quad 010 \mapsto 010, \quad 011 \leftrightarrow 101,$$
$$100 \mapsto 100, \quad 110 \mapsto 010, \quad 111 \mapsto 111.$$

In every cycle except $011 \leftrightarrow 101$, trace parity is constant. Oddness of q_r would then force s_r to be constant, contrary to Lemma 3.1. Thus

$$s_{r+1} = 1 - s_r \tag{8}$$

for every sufficiently large r.

Fix a large prime $q_r \nmid 3c\,\mathrm{disc}(f)$. If $v_3(q_r^d - 1) \le r$, the return proof with an even period gives $q_r \mid T_{r+2jK} - s_r = q_{r+2jK}$, a contradiction to eventual primality. Therefore

$$v_3(q_r^d - 1) \ge r + 1.$$

For $d = 1$ this forces $q_r \equiv 1 \pmod{3^{r+1}}$. For $d = 2$ it forces $q_r \equiv \pm 1 \pmod{3^{r+1}}$. For $d = 3$, it first forces $q_r \equiv 1 \pmod 3$, and the lifting-the-exponent identity gives $v_3(q_r^3 - 1) = v_3(q_r - 1) + 1$, so $q_r \equiv 1 \pmod{3^r}$. Uniformly,

$$q_r \equiv \varepsilon_r \pmod{3^r}, \qquad \varepsilon_r \in \{-1, 1\}.$$

By (7), the sign is uniquely $\varepsilon_r = 2s_r - 1$. Hence

$$T_r \equiv 3s_r - 1 \pmod{3^r}.$$

The representatives on the right alternate between -1 and 2. Their difference is ± 3, whereas (5) requires $T_{r+1} - T_r \equiv 0 \pmod{3^r}$. For $r \ge 2$ this is impossible. Theorem 1.1 follows.

4 Real shifts and a backward 3-adic obstruction

Proposition 4.1 (Prime-tail rigidity; [source T34,36]). *For a totally real cubic Pisot number with shift s, if $T_r - s$ is eventually prime, there exists a fixed $\varepsilon \in \{-1, 1\}$ such that*

$$T_r - s \equiv \varepsilon \pmod{3^{r+1}}$$

for every sufficiently large r. Consequently $L_f = s + \varepsilon$, and

$$T_k \equiv s + \varepsilon \pmod{3^{k+1}} \quad \text{for every } k \geq 0.$$

When $s = 0$, also $3^{r+2} \mid T_{r+1} - T_r^3$ for large r.

Proof. The affine-return contradiction used above first gives $T_r - s \equiv \pm 1 \pmod{3^r}$. The residue modulo 3 fixes the sign independently of r. Apply this at index $r+1$ and bring the congruence back using (5); this raises the modulus for index r to 3^{r+1}. Letting r tend to infinity gives $L_f = s + \varepsilon$, and (6) gives every earlier congruence. For $s = 0$, cubing $T_r \equiv \varepsilon \pmod{3^{r+1}}$ gives $T_r^3 \equiv \varepsilon \pmod{3^{r+2}}$, also satisfied by T_{r+1}. □

We make the coefficient consequences entirely explicit.

Lemma 4.2 (Residue dependence). *The limit L_f depends only on $(a, b, c) \bmod 3$. It is the sum, with multiplicities, of the Teichmüller representatives of the nonzero roots of $f \bmod 3$ in an algebraic closure of $\mathbb{F}_3$.*

Proof. Work in a finite extension of $\mathbb{Q}_3$ containing all roots. A root of positive valuation has 3^rth power tending to zero. For a unit root x, write $x = \omega(x)u$, where $\omega(x)$ is its prime-to-3 root-of-unity representative and u is a principal unit. Then $u^{3^r} \to 1$. Frobenius permutes the residual roots with their multiplicities, so $\sum \omega(x)^{3^r} = \sum \omega(x)$ for every r. Passing to the limit proves the assertion. This also covers repeated residual roots and ramified splitting fields. □

Proposition 4.3 (Exact small-limit classification; [source T36,50]). *For $\varepsilon = \pm 1$,*

$$L_f = \varepsilon \iff a \equiv \varepsilon \pmod 3 \text{ and } c \equiv ab \pmod 3.$$

Moreover,

$$L_f = 0 \iff a \equiv c \equiv 0 \pmod 3, \quad L_f = 2 \iff (a, b, c) \equiv (2, 1, 0) \pmod 3.$$

Proof. If $L_f = \varepsilon$, the congruences at $k = 0, 1$ give $a \equiv \varepsilon \pmod 3$ and $a^3 - 3ab + 3c \equiv \varepsilon \pmod 9$. Since $a^3 \equiv \varepsilon \pmod 9$, necessarily $c \equiv ab \pmod 3$. Conversely this factorizes the residual polynomial as $(X - \varepsilon)(X^2 + b)$. The two roots of the quadratic are either zero or an opposite pair; their Teichmüller representatives sum to zero. Lemma 4.2 gives $L_f = \varepsilon$.

For the two remaining assertions, the following complete residue computation supplies necessary conditions. Each entry is $L_f \bmod 81$; rows are $b = 0, 1, 2$ and columns are $c = 0, 1, 2$:

$a=0$	0	1	2	$a=1$	0	1	2	$a=2$	0	1	2
$b=0$	0	3	78	$b=0$	1	58	7	$b=0$	80	74	23
$b=1$	0	21	60	$b=1$	79	1	4	$b=1$	2	77	80
$b=2$	0	39	42	$b=2$	22	52	1	$b=2$	59	80	29

These 27 entries are obtained by applying (3) three times modulo 81 to each representative $(a, b, c) \in \{0, 1, 2\}^3$: (6) fixes $T_3 \bmod 81$. Lemma 4.2 justifies using these representatives for arbitrary integral lifts. For sufficiency, the first surviving polynomial is $X(X^2 + b)$, whose lifted root sum is zero. The second is $X(X - 1)^2 \bmod 3$, whose lifted root sum is two. □

If $s = 0$ and $c \not\equiv ab \pmod 3$, Proposition 4.3 contradicts Proposition 4.1. If $s = 1$, a prime tail forces $L_f \in \{0, 2\}$, hence $3 \mid c$. Oddness of $T_r - 1$ requires $T_r \equiv 0 \pmod 2$ eventually. The state graph in the preceding section shows that no cycle with $E \equiv 1 \pmod 2$ has trace permanently zero. Thus $2 \mid c$ also. This proves Theorem 1.2.

Example 4.4 (A full infinite family)**.** For every integer $A \geq 4$, let β_A be the large root of $X^3 - AX^2 + 2$. The values at $-1, 0, 1, A-1, A$ locate the three roots in $(-1, 0)$, $(0, 1)$, and $(A-1, A)$. None of the possible rational roots $\pm 1, \pm 2$ occurs. Thus β_A is cubic Pisot and the small-root sum $A - \beta_A$ is positive. Here $c = -2$, so infinitely many $\lfloor \beta_A^{3^r} \rfloor$ are composite. This argument covers the entire indicated family, not merely sampled values of A.

5 Positive-density obstructions and an anti-cascade

For a set $S \subseteq \mathbb{Z}_{\geq 0}$, write $\underline{d}(S) = \liminf_{R\to\infty} |S \cap [0, R)|/R$.

Proposition 5.1 (Order-budget criterion; [source T46])**.** *For totally real cubic Pisot β, put $z = L_f - s$. If $z \notin \{-1, 1\}$, the composite indices have positive lower density. The same holds when $z = -1$ and* $\mathrm{disc}(f)$ *is a square in $\mathbb{Z}$.*

Proof. If $z \neq \pm 1$, both $v_3((T_r - s) - 1)$ and $v_3((T_r - s) + 1)$ are bounded for large r. At any sufficiently late prime term $q = T_r - s$, the quantity $v_3(q^d - 1)$ is bounded uniformly for $d = 1, 2, 3$ by the lifting-the-exponent formulas. Choosing r beyond this bound gives an arithmetic progression of composite terms by Lemma 2.3. If no such late prime term exists, the whole tail is composite.

If $z = -1$, then (6) implies $q \equiv -1 \pmod 3$ for large r. For unramified primes, square discriminant excludes a transposition of roots, so only splitting degrees $d = 1$ and $d = 3$ occur. In either case $3 \nmid q^d - 1$, and the same return argument works. The finitely many ramified primes cannot equal an unbounded late prime term. □

Example 5.2 (A prime seed that never returns; [source T46])**.** Let $f = X^3 - 7X^2 - 2X + 2$. Its roots lie in $(-1, 0)$, $(0, 1)$, and $(7, 8)$, and the rational-root test gives irreducibility. Its small-root sum is negative, so all odd powers have floor equal to trace. The first two states are

$$(T_0, B_0, E_0) = (7, -2, -2), \qquad (T_1, B_1, E_1) = (379, -80, -8).$$

The number 379 is prime by division by primes at most 19. Modulo 379, the state after index one passes through

r	T_r	B_r	E_r
2	355	221	246
3	173	367	195
4	189	307	119
5	12	136	125
6	239	376	138
7	251	336	86
8	277	370	94
9	173	367	195

Each row follows by direct substitution into (3); row nine repeats row three. No subsequent trace coordinate is zero. Thus 379 divides none of the later floors. This disproves a general seed-propagation principle, but not Saito's problem: the next floor is $54\,530\,875 = 5^3 \cdot 7^2 \cdot 29 \cdot 307$.

6 A quadratic-character cycle sieve

Assume $s = 0$ and $L_f = -1$. If the discriminant is not a rational square, let D be the fundamental discriminant of $\mathbb{Q}(\sqrt{\operatorname{disc}(f)})$, and let χ_D denote its quadratic character.

Theorem 6.1 (Resolvent cycle; [source T50])**.** *Iterate (3) modulo $|D|$. If its eventual cycle contains a state with $\chi_D(T) \neq -1$, the composite indices have positive lower density.*

Proof. Every cycle position corresponds to an infinite arithmetic progression of indices. If $\chi_D(T) = 0$, fix a prime divisor of $\gcd(T, D)$ at that position; it divides every later trace at that position, proving the assertion.

If $\chi_D(T) = 1$, either all sufficiently late terms on that progression are composite, or it contains an arbitrarily late prime seed $q = T_r$ not dividing $3c\operatorname{disc}(f)$. The Frobenius permutation at q is even: its sign is the quadratic character of the discriminant. Thus the factorization type is $1+1+1$ or 3, with splitting degree one or three. As $L_f = -1$, one has $q \equiv -1 \pmod 3$, so $3 \nmid q^d - 1$. Lemma 2.3 gives a composite progression. Both alternatives have positive lower density. □

Example 6.2. For $f = X^3 - 8X^2 - 2X + 1$, the sign tests as above give $s = 0$. The discriminant is $2597 = 7^2 \cdot 53$ and $c - ab = 15$, so $L_f = -1$ and $D = 53$. The complete modulo-53 state cycle is

$$(8, 51, 52), (27, 0, 52), (17, 3, 52), (40, 24, 52), (8, 12, 52), (9, 5, 52).$$

The characters of the trace coordinates are respectively $-1, -1, 1, 1, -1, 1$. Theorem 6.1 therefore applies. This example is outside both the coefficient obstruction $c \not\equiv ab \pmod 3$ and the square-discriminant criterion.

7 Projective return and Frobenius restrictions

Theorem 7.1 (Projective trace-zero return; [source T51])**.** *Let $p \nmid 3c\operatorname{disc}(f)$ divide T_r. In a splitting field modulo p, put $Y_i = \alpha_i^{3^r}$. Let m be the exponent of the subgroup generated by all Y_i/Y_j. If $(m, 3) = 1$, there exists $K \geq 1$ such that*

$$p \mid T_{r+jK} \qquad (j \geq 0).$$

For a real Pisot number with zero shift this produces a composite progression. The condition on m holds automatically in either case:

(i) $r \geq 1$ and $f \bmod p$ is irreducible;

(ii) $f \bmod p$ has factorization type $1 + 2$, $v_3(p-1) = r + 1$, and c is a cube modulo p.

Proof. Take $K = \operatorname{ord}_m(3)$, with the usual convention for $m = 1$. Then $(Y_i/Y_1)^{3^{jK}} = Y_i/Y_1$ for every i, j. Thus a common nonzero scalar λ_j satisfies $Y_i^{3^{jK}} = \lambda_j Y_i$ for all i. Their sum remains zero.

In the irreducible case the roots have the form x, x^p, x^{p^2} in $\mathbb{F}_{p^3}$. Each ratio has norm one down to $\mathbb{F}_p$, so its order divides $p^2 + p + 1$. For $p \neq 3$ this integer has 3-adic valuation at most one. Raising all roots to 3^r with $r \geq 1$ removes that part.

For type $1 + 2$, write the roots as x, y, y^p, with $x \in \mathbb{F}_p$ and $y \in \mathbb{F}_{p^2}$. The hypothesis $v_3(p-1) = r + 1$ implies $p \equiv 1 \pmod 3$ and $3 \nmid p + 1$. Hence, after powering by 3^r, the 3-primary components of the roots are $\eta_1, \eta_2, \eta_2 \in \mu_3$. The product is c^{3^r}. Since c is a cube in $\mathbb{F}_p$, its 3-primary component is annihilated by 3^r. Therefore $\eta_1 \eta_2^2 = 1$, which in μ_3 is equivalent to $\eta_1 = \eta_2$. All ratios have trivial 3-primary part. □

The last proof uses zero trace essentially. A common scalar preserves $\sum Y_i = 0$; it does not in general preserve $\sum Y_i = s$ when $s \neq 0$. Nor does irreducibility of the original rational polynomial imply irreducibility modulo a given trace divisor.

8 Coefficient families and explicit covering progressions

For $A \geq 4$ set

$$f_A^-(X) = X^3 - AX^2 - 2X + 1, \qquad f_A^+(X) = X^3 - AX^2 - 2X + 2. \tag{9}$$

Each polynomial has one root in each of $(-1, 0), (0, 1), (A, A+1)$, by evaluating at the endpoints. The rational-root test excludes ± 1 for f_A^- and $\pm 1, \pm 2$ for f_A^+. Their large roots are therefore cubic Pisot. If the small roots are $u < 0 < v$, their sum is $A - \beta < 0$, so $|u| > v$ and

$$-1 < u^N + v^N < 0 \qquad (N \text{ positive and odd}).$$

Consequently floor and trace agree at every index considered below.

Theorem 8.1 (Density-one divisor families; [source T52])**.** *For f_A^-, suppose A has a prime divisor $p \equiv 11$ or $14 \pmod{15}$. For f_A^+, suppose it has an odd prime divisor $p \equiv 2 \pmod 3$ with $\left(\frac{-19}{p}\right) = 1$. In either case the corresponding composite indices have positive lower density. Each sufficient coefficient condition holds for a set of A of natural density one. It also has relative density one within, respectively, $A \equiv 2 \pmod 6$ and $A \equiv 1 \pmod 6$.*

Proof. The discriminants are

$$\Delta_A^- = 4A^3 + 4A^2 + 36A + 5, \qquad \Delta_A^+ = 4(2A^3 + A^2 + 18A - 19).$$

For $p \mid A$, these reduce to 5 and -76. In the first case the stated residue classes give $(5/p) = 1$ and $p \equiv -1 \pmod 3$; in the second these facts follow from the hypotheses. Both discriminants are nonzero squares modulo p, and $p \nmid c$. Thus the splitting degree is $d = 1$ or 3, and $3 \nmid p^d - 1$. Since $p \mid T_0 = A$, affine return applies already at index zero.

The set of good primes in the first case is a union of two reduced classes modulo 15. In the second it is a nonempty union of reduced classes modulo 57; for example $p \equiv 5 \pmod{57}$ is good. Dirichlet's theorem with its standard reciprocal-prime-divergence consequence implies that each good-prime set $\mathcal{P}$ has $\sum_{p \in \mathcal{P}} 1/p = \infty$. For any finite subset F, the integers avoiding divisibility by every $p \in F$ have density $\prod_{p \in F}(1 - 1/p)$. Letting F increase shows upper density zero for integers with no good prime divisor. The same argument inside either progression modulo 6 uses the Chinese remainder theorem; all good primes are coprime to 6. □

The residue classes mentioned in the last sentence are the ones surviving the simplest parity and 3-adic tests for these families. The density assertion refers to the coefficient parameter A, not to all cubic Pisot numbers and not to hypothetical Mills constants. In particular a coefficient detected through a proper divisor is composite from its first floor.

Theorem 8.2 (Prime-coefficient covers; [source T52])**.** *If $A \equiv 367 \pmod{510}$, then the large root of f_A^+ satisfies*

$$\left\lfloor \beta^{3^r} \right\rfloor \text{ composite for every } r \geq 1.$$

There are infinitely many prime such A, and then $\lfloor \beta \rfloor = A$ is prime. Separately, for a prime $A \equiv 11 \pmod{42}$, let β be the large root of

$$g_A(X) = X^3 - AX^2 - X + 5.$$

Then $7 \mid \left\lfloor \beta^{3^{3+4j}} \right\rfloor$ for every $j \geq 0$.

Proof. For f_A^+, the initial states modulo 5 and 17 and their exact cycles are

$$(2,3,3) \mapsto (4,0,2) \mapsto (0,2,3) \mapsto (4,0,2) \pmod 5,$$
$$(10,15,15) \leftrightarrow (0,3,9) \pmod{17}.$$

Thus $5 \mid T_{2j}$ for $j \geq 1$ and $17 \mid T_{2j+1}$ for $j \geq 0$. The traces exceed their displayed divisors. Since $\gcd(367, 510) = 1$, Dirichlet's theorem supplies infinitely many prime coefficients.

For g_A, the prime coefficient is at least 11. Endpoint signs put the roots in $(-1,0), (0,1), (A, A+1)$; the only possible rational roots $\pm 1, \pm 5$ are excluded. The negative small root dominates in modulus, giving zero shift for all odd powers. Modulo 7 the initial state and orbit are

$$(4,6,2) \mapsto (5,0,1) \mapsto (2,3,1) \mapsto (0,5,1) \mapsto (3,2,1) \mapsto (5,0,1).$$

This proves the asserted progression. Again Dirichlet's theorem applies because $\gcd(11, 42) = 1$. □

These families also reach the small-conjugate region relevant to Saito's Mills reduction. For f_A^+ and $A \geq 16$, write the small roots as $-\rho, \sigma$. Their defining equations give

$$A^{-1/2} < \sigma < \rho < 2A^{-1/2}.$$

For example, $\sigma < 1/2$ follows from $f_A^+(1/2) < 0$, and then $A\sigma^2 = \sigma^3 - 2\sigma + 2 > 1$; the bound on ρ follows from $A\rho^2 = 2 + 2\rho - \rho^3 < 4$. It follows that

$$\rho \leq \min(\sigma^{17/23}, \beta^{-17/40})$$

whenever A is sufficiently large: it suffices that $A^3 \geq 2^{23}$ and $A^{20} \geq 2^{40}(A+1)^{17}$. For g_A, similarly $\rho < \sqrt{6/A}$ and $\sigma > 2/\sqrt{A}$; sufficient conditions are

$$A^6 2^{11} \geq 3^{23}, \qquad A^{20} \geq 6^{20}(A+1)^{17}.$$

Thus the progression arguments do not rely on large small-conjugates that are already excluded by Saito's bounds. They still remove only particular families.

9 Remaining cases and proof-status summary

The real prime-tail conditions reduce to $L_f - s \in \{-1, 1\}$. In the zero-shift branch, the character sieve can fail when its whole cycle has value -1. Projective return removes irreducible cubic reduction but leaves completely split reductions with residual 3-primary ratio orders. Type $1+2$ can also survive when the 3-adic depth is larger than the exact one-place criterion, or when the norm has the wrong cubic residue. These are genuine gaps in coverage, not exceptional cases already proved absent.

Saito's Mills reduction [2] leaves a totally real cubic Pisot power with a dominant negative small conjugate and additional size bounds. The non-real theorem therefore does not by itself settle Mills' transcendence. The coefficient and projective tests provide possible exclusion certificates for real candidates; a theorem that forces such a certificate for every candidate is still missing.

Source	Result included here
T34, T36	Prime-tail congruences, backward limit, real coefficient obstructions.
T45	All non-real cubic Pisot numbers: infinitely many composite terms.
T46	Positive-density order test and a complete counterexample to seed propagation.
T50	Exact small-limit classification and a resolvent-cycle sieve.
T51	Affine divisor return and projective trace-zero return.
T52	Density-one coefficient families, prime-coefficient covers, certified prefix below.

A A reproducible three-prime prefix

For $A = 171917$ in g_A, the first three traces are

$$\begin{aligned} T_0 &= 171917, \\ T_1 &= 5081085138667949, \\ T_2 &= 131180540514334609193223651727726951788970456781. \end{aligned}$$

They are all prime; meanwhile $7 \mid T_3$ by Theorem 8.2. To make the primality claim independent of any missing computational attachment, we supply a finite recursive certificate below.

The criterion used is elementary: if the complete prime factorization $N - 1 = \prod q^{e_q}$ is known and some integer w satisfies

$$w^{N-1} \equiv 1 \pmod{N}, \qquad \gcd(w^{(N-1)/q} - 1, N) = 1 \quad \text{for every } q \mid N - 1,$$

then N is prime. Indeed, for every prime $\ell \mid N$, the order of $w \bmod \ell$ is divisible by every q^{e_q}, so $N - 1 \mid \ell - 1$ and $\ell \geq N$. The table records the full factorizations and a witness; all leaf primes below 100 are certified by trial division. Every displayed congruence is checked by repeated squaring, so the table is a finite exact certificate rather than a probabilistic primality test.

i	q_i	witness w_i
1	101	2
2	109	6
3	379	2
4	401	3
5	1153	5
6	1973	2
7	15233	3
8	18181	2
9	42979	3
10	117041	3
11	136949	3
12	171917	2
13	936329	3
14	968521	19
15	2589611	2
16	67329887	5
17	95295139957	2
18	1492680710537	3
19	1876079971369	11

i	q_i	witness w_i
20	534986915718599	7
21	5081085138667949	2
22	506042170576387748411234 57	3
23	1898751161772912282533582654647086052660 1	3
24	1311805405143346091932236517277269517889704567 81	2

i	Factorization of $q_i - 1$
1	$2^2 \cdot 5^2$
2	$2^2 \cdot 3^3$
3	$2 \cdot 3^3 \cdot 7$
4	$2^4 \cdot 5^2$
5	$2^7 \cdot 3^2$
6	$2^2 \cdot 17 \cdot 29$
7	$2^7 \cdot 7 \cdot 17$
8	$2^2 \cdot 3^2 \cdot 5 \cdot q_1$
9	$2 \cdot 3 \cdot 13 \cdot 19 \cdot 29$
10	$2^4 \cdot 5 \cdot 7 \cdot 11 \cdot 19$
11	$2^2 \cdot 7 \cdot 67 \cdot 73$
12	$2^2 \cdot q_9$
13	$2^3 \cdot q_{10}$
14	$2^3 \cdot 3 \cdot 5 \cdot 7 \cdot q_5$
15	$2 \cdot 5 \cdot 17 \cdot q_7$
16	$2 \cdot 13 \cdot q_{15}$
17	$2^2 \cdot 3^3 \cdot 17 \cdot q_3 \cdot q_{11}$
18	$2^3 \cdot q_1 \cdot q_6 \cdot q_{13}$
19	$2^3 \cdot 3^4 \cdot 43 \cdot q_{16}$
20	$2 \cdot 7 \cdot q_4 \cdot q_{17}$
21	$2^2 \cdot 23 \cdot 37 \cdot q_{18}$
22	$2^7 \cdot 7 \cdot q_2 \cdot q_{14} \cdot q_{20}$
23	$2^3 \cdot 5^2 \cdot q_{19} \cdot q_{22}$
24	$2^2 \cdot 5 \cdot 19 \cdot q_8 \cdot q_{23}$

References

[1] K. Saito, *Intervals without primes near an iterated linear recurrence sequence*, arXiv:2504.14968v1 (2025); International Journal of Number Theory (2026), doi:10.1142/S1793042126500685. Problem numbering here follows the arXiv version.

[2] K. Saito, *Transcendency of variants of Mills' constant*, arXiv:2508.16068v3 (7 December 2025), especially Theorem 1.7 and Lemmas 5.7, 5.10; https://arxiv.org/abs/2508.16068.

[3] K. Saito, *Mills' constant is irrational*, Mathematika **71** (2025), e70027; https://arxiv.org/abs/2404.19461.

[4] A. Dubickas, *Integer parts of powers of Pisot and Salem numbers*, Archiv der Mathematik **79** (2002), 252–257. Referenced through the explicit statements and proofs in [2].